

Deep Learning Framework for Smart Paddy Health Analysis and Yield Forecasting

AI-Based Precision Agriculture System for Early Disease Detection and Production Optimization

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Abstract—Agriculture plays a vital role in global food security, with rice being one of the most important staple crops worldwide. Paddy cultivation is highly affected by plant diseases and environmental factors, which directly impact crop yield and quality. Early disease detection and accurate yield forecasting are essential for improving productivity and ensuring sustainable farming practices.

This paper proposes a deep learning–based framework for smart paddy health analysis and yield prediction. Convolutional Neural Networks (CNNs) are used to detect diseases from leaf images at early stages, while regression models utilize environmental and historical data to forecast crop yield. The system integrates image-based disease diagnosis with data-driven yield estimation to provide timely recommendations for farmers.

Experimental results show high classification accuracy and reliable yield prediction performance. The proposed framework supports precision agriculture by enabling early intervention, optimized resource management, and improved agricultural productivity.

Index Terms— Deep Learning, Paddy Disease Detection, Yield Forecasting, Convolutional Neural Network, Precision Agriculture.

1.INTRODUCTION

Agriculture plays a crucial role in ensuring food security and economic stability worldwide. Paddy cultivation, in particular, is highly sensitive to plant diseases, climatic conditions, and soil variability, all of which significantly affect crop productivity and quality. Early detection of diseases and accurate yield estimation are essential to minimize losses and improve farm management. However, traditional disease identification methods rely on manual inspection by agricultural experts, which is time-consuming, labor-intensive, costly, and prone to human error. Similarly, conventional yield estimation techniques often fail to capture complex interactions among environmental, soil, and historical factors.

With the rapid advancement of Artificial Intelligence (AI) and Deep Learning, automated image analysis and predictive modeling have emerged as powerful tools in smart agriculture. Convolutional Neural Networks (CNNs) have demonstrated high performance in plant disease classification, while machine learning regression models have proven effective for yield forecasting.

This paper proposes a unified intelligent agricultural system for smart paddy health monitoring and yield prediction. As illustrated in the system architecture, the framework consists of a user interface for uploading leaf images, a preprocessing module for image enhancement, a CNN-based feature extraction and disease prediction module, and a severity assessment component to evaluate the level of infection. A recommendation system then provides disease information and treatment guidance, which can be translated into regional languages for better accessibility. The backend connects the frontend, model, and database to ensure seamless data flow and storage.

In addition to disease detection, the system integrates environmental and historical agricultural data to forecast crop yield using regression models. By combining image-based health analysis with data-driven prediction, the proposed system delivers accurate, real-time insights to farmers, enabling timely intervention, optimized resource utilization, and improved agricultural productivity.